

The project consists of creating a Brick Breaker game developed in Fortran. The player controls a paddle that moves horizontally to bounce a ball and destroy the bricks located at the top of the screen.

User Guide :

When launching the program, a first prompt asks the user to enter the number of rows for the grid, then a second prompt asks them to specify the number of columns.

To control the paddle, the commands are as follows:

- **Move left** : key **g**
- **Move right** : key **h**

The 'X' bricks are standard bricks, while the '2' bricks are special bricks.

Issues:

- **Difficulty:** We noticed that while the ball was moving, it would sometimes change direction even though there was no obstacle. We realized that some bricks were invisible but actually present. **Solution:** We had to reset the bricks so that they wouldn't appear in the lower part of the grid.
- **Difficulty:** We faced an issue related to our coordinate system, which is inverted: "x" represents the ordinate (vertical axis) and "y" the abscissa (horizontal axis). This was unintuitive and caused several inconsistencies. **Solution:** Instead of changing all the notations throughout the code, we adapted to this new coordinate system.
- **Difficulty:** The movement of the ball was constrained to change direction one space before the wall/boundary. However, in our initial brick setup, some bricks were placed one space before the boundary. When the ball hit one of these bricks, it faced a double constraint, which caused the ball's movement to glitch. **Solution:** We decided to remove the column of bricks adjacent to the boundaries to avoid this double constraint.

Improvements:

There are special bricks marked '2'. When the ball hits one of these bricks, the paddle shrinks by one unit in width.